Systems Engineering Plan

SQRT (Sub-zero Quantitative Research Team)

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Key Reference Documents

1. SQT-PDR-SEP-0
2. SQT-CDR-PMP-1
3. SQT-CDR-SDF-0
4. UCD-DRD-2026-0

1 Executive Summary

This document is a revised version of the Systems Engineering Plan from the preliminary design review. Revisions will be made where necessary to reflect the current mission status and workflow. All revisions made will be detailed in red text at the beginning of the relevant section.

A brief mission overview will be given, outlining the main mission objectives and the systems that will be required to fulfill them. This will lead into the definition of the refined, finalised mission requirements in section 3.

In section 4, the processes necessary to satisfy these mission & technical requirements will be presented visually via function tree. While the function tree will not have evolved substantially since PDR, the resources required to perform them have. These products will be found in the product tree in section 5.

Finally, the introduction of an interface control plan will describe how the subsystems comprising the on-board segment will operate together. In order to maximise mission returns and minimize risk, subsystem operating conditions and requisites will need to be considered, preventing any potential conflicts.

2 Mission Overview

Mission Statement: The SQRT mission will produce and observe the flash evaporation of water via exposure to a low pressure and low temperature environment to investigate how such plumes thermally evolve.

The flash evaporation process is relevant to cryovolcanic plumes on icy moons in our solar system as well as desalinisation plants on Earth. In order to reproduce the conditions required for flash evaporation, the isolated water container will be flown to an altitudes in excess of 20 km, where temperatures and pressures are lower. Upon sudden exposure to this environment, it is expected that the water will undergo a near-instantaneous phase transition. The mission objectives are concerned with observing this process and profiling the thermal evolution of the water sample throughout.

2.1 Mission Goals

The success of the SQRT mission will be determined by a number of metrics. Firstly, the flash evaporation event will have to be detected and verified. In this phase, the current idea is to perform direct observation via use of a thermal sensor. The rapid phase transition will hopefully be decipherable and interpretable via analysis of the thermal evolution of the water. The frames acquired by this sensor will be stored onboard the TupperSat by means of an SD card. In the event of non-retrieval of the TupperSat, downsampled frames will be downlinked for use during mission validation.

A preliminary metric of mission success will be the activation of the exposure mechanism and its deployment. Satisfying of the metric will done using a temperature sensor could be placed near the valve to confirm a rise in temperature due to the introduction of the relatively warm water. Successful activation of the pressure trigger sensor at the chosen target pressure (35 mbar), or by other secondary conditions, will be signified by switching of a flag to be transmitted.

Fulfilling these criteria and maximising mission return will require the design of a lightweight system that is economical with data downlink and is compartmentalised to prevent water damage to electronics. The assembly of this system in phase 3 will be guided by the mission and technical requirements outlined in this document.

3 Mission Requirements

Changes to requirements are shown in red.

1. SQT:MR:020 - Pressure Data: A pressure sensor may not be required to determine the pressure in the low-pressure chamber. Vents should ensure that pressure is normalised to external pressure levels. The requirement was removed for CDR.

2. SQT:MR:060 - The specific thermal array is now known, as is its refresh rate.

3. SQT:MR:100 - Water Exposure: The exposure altitude has been reduced to approx. 23 km to reduce the risk of balloon bursting before trigger condition is met.

4. SQT:TR:00400 & SQT:TR:00410 - Chamber Pressure Sensor: These requirement were also removed as there are no plans to house a pressure sensor in the low-pressure chamber. They were replaced with requirements related to the internal temperature sensor

All mentions of "thermal camera" changed to "thermal sensor"

The mission requirements stipulated below are divided into a number of sections. Level 0 requirements correspond to the mission objectives, against which the success of the mission will be measured. Level

1 requirements are high-level mission requirements directly linked to the mission objective, data acquisition, and retrieval. The level 2 requirements are technical requirements concerned with the general operation of the TupperSat platform and structure. These requirements will also be documented and traced by means of a requirement traceability matrix during the successive phases of the mission.

3.1 Level 0: Mission Objectives

MO1 Fly and recover a TupperSat containing an experimental scientific or engineering/technology demonstration payload with potential application to space science.

MO2 SQRT will attempt to reproduce and observe the flash evaporation process of water exposed to near-space conditions.

3.2 Level 1: Mission Requirements

3.2.1 Customer (UCD) Requirements

Table 1: Customer (UCD) Mission Requirements

Requirement ID	Title	Description	Justification
UCD:MR:0010	Payload Size	The TupperSat shall be carried as a payload on a light unmanned free balloon.	Inherited from UCD-PRD-2026-0.
UCD:MR:0020	Flight Time	The TupperSat shall operate for at least 6 hours.	Inherited from UCD-PRD-2026-0.
UCD:MR:0030	Flight Environment	The TupperSat shall operate up to 35 km altitude.	Inherited from UCD-PRD-2026-0.
UCD:MR:0040	TDRS Communication	The TupperSat shall communicate with a ground station using TDRS.	Inherited from UCD-PRD-2026-0.
UCD:MR:0050	Tracking and Recovery	The TupperSat shall report real-time position telemetry.	Inherited from UCD-PRD-2026-0.
UCD:MR:0060	Non-Recovery	The TupperSat shall transmit any experimental data necessary to meet the mission objectives in the event that recovery is not possible.	Inherited from UCD-PRD-2026-0.
UCD:MR:0070	Atmospheric Profiling	The TupperSat shall measure and report the atmosphere's temperature and pressure profiled with vertical resolution of at least 1 measurement per 200 m ascent (goal: at least 1 measurement per 100 m).	Inherited from UCD-PRD-2026-0.
UCD:MR:0080	Launch Readiness	The TupperSat shall be deilvered ready to be turned on for launch without further intervention.	Inherited from UCD-PRD-2026-0.

3.2.2 Mission Specific Requirements

Table 2: Mission Specific Requirements

Requirement ID	Title	Description	Justification
SQT:MR:010	Thermal Data	The TupperSat shall measure the temperature of the water and/or surrounding vapour during exposure to reduced pressure	To characterise and quantify its thermal evolution and identify key transition events.
SQT:MR:030	Timestamped Data	The TupperSat shall record the time at each primary measurement.	Storing timestamps at each data acquisition will facilitate chronological results.
SQT:MR:040	Altitude Data	The TupperSat shall record the altitude for all primary data acquisitions.	Recording the altitude will verify if the trigger sensor and exposure mechanism functioned at the desired altitude.
SQT:MR:050	Trigger Operation	The TupperSat shall provide verifiable confirmation that the trigger condition has been met	This requirement supports UCD:MR:0060 for the scenario of non-recovery of the TupperSat.
SQT:MR:060	Exposed Sample Containment	The TupperSat shall maintain the water sample within the low-pressure chamber for at least 2 seconds	This requirement allows sufficient time for the primary data to be acquired.
SQT:MR:070	External Environment Readings	The TupperSat shall record continuous external pressure data during the exposure period.	This requirement will aid post-mission analysis. External pressure can be compared to the pressures inside the low-pressure chamber to appraise performance of venting.
SQT:MR:080	Ease of Data Retrieval	The TupperSat shall be designed in such a way that primary experimental data is recorded, stored and retrievable following payload recovery	For post-mission analysis and subsequent confirmation of mission success. This process will be covered in the DMP.
SQT:MR:090	Viability for Prototyping	The TupperSat shall be designed to allow repeatable ground-based testing	To ensure reliability and durability of mission-critical subsystems.
SQT:MR:100	Water Exposure	The TupperSat shall be designed in such a way to expose liquid at atmospheric pressure of $\leq$ 35 mbar at approx. 23 km altitude.	This requirement helps to satisfy MO2 and to ensure that flash evaporation occurs.

3.3 Level 2: Technical Requirements

3.3.1 Customer (UCD) Requirements

3.3.1.1 Structure

Table 3: Structural Requirements

Requirement ID	Title	Description	Justification
UCD:TR:00010	TupperSat Mass	The TupperSat's total mass shall not exceed 0.8 kg.	Inherited from UCD-PRD-2026-0.
UCD:TR:00020	TupperSat Volume	The TupperSat shall be enclosed in a Tupperware-like container, with volume less than 2 litres.	Inherited from UCD-PRD-2026-0.
UCD:TR:00030	Launch Interface	The TupperSat shall attach to the launch balloon using a single carabiner connector.	Inherited from UCD-PRD-2026-0.

3.3.1.2 - Communications

Table 4: Communications Requirements

Requirement ID	Title	Description	Justification
UCD:TR:00100	Packet Transmission Frequency	The TupperSat shall transmit packets at 868 MHz.	Inherited from UCD-PRD-2026-0.
UCD:TR:00110	Packet Drop Rate	The TupperSat shall achieve a packet drop rate of less than 20% transmitting over a distance of at least 12 km.	Inherited from UCD-PRD-2026-0.
UCD:TR:00120	Telemetry Packet Rate	The TupperSat shall transmit at least 2 and no more than 3 housekeeping telemetry packets per minute throughout the flight duration.	Inherited from UCD-PRD-2026-0.
UCD:TR:00130	Data Packet Rate	The TupperSat shall transmit no more than 3 data packets per minute throughout the flight duration.	Inherited from UCD-PRD-2026-0.

3.3.1.3 Housekeeping Telemetry & Atmospheric Data

Table 5: Housekeeping Telemetry & Atmospheric Data

Requirement ID	Title	Description	Justification
UCD:TR:00200	Telemetry - Altitude	The TupperSat shall record its altitude at least every 10 s with accuracy better than $\pm$ 50 m.	Inherited from UCD-PRD-2026-0.
UCD:TR:00210	Telemetry - Latitude & Longitude	The TupperSat shall record horizontal position (latitude and longitude) with horizontal accuracy better than 50 m.	Inherited from UCD-PRD-2026-0.

Requirement ID	Title	Description	Justification
UCD:TR:00220	Telemetry - Internal Temperature	The TupperSat shall record internal temperature with accuracy better than $\pm 2^{\circ}C$ at least every 20 s.	Inherited from UCD-PRD-2026-0.
UCD:TR:00230	Telemetry - External Temperature	The TupperSat shall record external temperature with accuracy better than $\pm 2^{\circ}C$ at least every 20 s.	Inherited from UCD-PRD-2026-0.
UCD:TR:00240	Telemetry - Atmospheric Pressure	The TupperSat shall record atmospheric pressure with accuracy better than 20 mbar at sea level at least every 20 s.	Inherited from UCD-PRD-2026-0.

3.3.1.4 Power

Table 6: Power Requirements

Requirement ID	Title	Description	Justification
UCD:TR:00300	TupperSat Power	The TupperSat battery system shall provide power for at least 6 hour of flight operations.	Inherited from UCD-PRD-2026-0.

3.3.1.5 Operations

Table 7: Operations Requirements

Requirement ID	Title	Description	Justification
UCD:TR:00400	Delivery	The TupperSat shall be delivered to the launch site ready to be launched by a single third-party operator in under 15 minutes	Inherited from UCD-PRD-2026-0 & quantifies UCD:MR:0080

3.3.2 Mission Specific Requirements

3.3.2.1 Water Container

Table 8: Water Container Requirements

Requirement ID	Title	Description	Justification
SQT:TR:00100	Water Temperature	The water container shall maintain the water at a temperature over $25^{\circ}C$.	At the threshold pressure of 35 mbar, water boils at around $27^{\circ}C$. Keeping the water close to this point will facilitate partial flash evaporation.
SQT:TR:00110	Pressure Resistant	The water container shall maintain structural integrity (no cracks or leaks) during ascent. This includes all pressure differentials that may be encountered.	This requirement ensures containment of water and prevention of sensor or electronic damage.

3.3.2.2 Exposure Mechanism

Table 9: Exposure Mechanism Requirements

Requirement ID	Title	Description	Justification
SQT:TR:00200	Valve Activation	The exposure mechanism shall send a signal after operation for downlink to ground segment.	This requirement will confirm that the exposure was executed. Satisfies SQT:MR:040 for event of non-recovery.
SQT:TR:00210	Leak Avoidance	The exposure mechanism shall prevent premature water exposure at environmental conditions up to the desired altitude.	This requirement is to prevent damage to electronics and to optimise data acquisition.
SQT:TR:00220	Valve Operation	The exposure mechanism shall operate at low-pressure chamber temperatures down to $-40°C$.	This requirement ensures that the exposure will occur, satisfying MO2 by allowing for the acquisition of meaningful thermal data.
SQT:TR:00230	Tubing Material	The tubing used for passing water into the chamber shall remain operable (allow water to pass through) at temperatures as low as $-40°C$.	Similarly to SQT:TR:00220, this requirement will ensure that water will reach the low-pressure chamber for thermal observations. Tubing will be more vulnerable than rest of exposure mechanism, justifying a dedicated requirement.
SQT:TR:00240	Valve/Servo Motor Compatibility	The valve shall be mechanically compatible with the selected servo motor throughout all mission phases (ascent, exposure, and descent).	This prevents any water leakage or catastrophic failures due to vibrations or sudden motions.

3.3.2.3 Thermal Sensor: Primary Data Acquisition

Table 10: Thermal Sensor Requirements

Requirement ID	Title	Description	Justification
SQT:TR:00300	Thermal Resolution	The thermal sensor shall have a thermal resolution of $\leq 0.1\,°\mathrm{C}$	This requirement helps to confirm the occurrence of a flash-evaporation event. The selected thermal resolution chosen is at least an order of magnitude smaller than the temperature change caused by evaporative cooling during flash evaporation.

Requirement ID	Title	Description	Justification
SQT:TR:00310	Sensor Function	Thermal sensor shall remain functional in temperatures down to -10 $^{\circ}C$.	This requirement will allow for measurement taking in ventilated, low-pressure chamber. If the sensor doesn't meet this requirement, it will have to be insulated or passively heated in order to remain functional.
SQT:TR:00320	Frame Rate	Thermal sensor shall have a frame rate of over 30 frames per second	Flash-evaporation occurs in less than a second, necessitating a sufficient frame rate to observe the process, as opposed to recording only pre- and post-event temperatures.
SQT:TR:00330	Calibration Stability	The thermal sensor shall maintain calibration accuracy over the expected flight temperature range.	Ensures validity of recorded thermal data.
SQT:TR:00340	Field-of-View	The thermal sensor shall have an FOV of at least 15 degrees.	This requirement ensures that, with current preliminary designs, the entire exposure mechanism can be seen by the thermal sensor. This supports SQT:MR:010.

3.3.2.4 Secondary Data Acquisition

Table 11: Sensors Requirements

Requirement ID	Title	Description	Justification
SQT:TR:00420	Temperature Sensor Sampling Rate	The temperature sensor positioned by the valve shall have a sufficient frame rate to capture significant rises in temperature linked to water movement.	This will help to confirm the successful function of the valve and the release of the water from the container.
SQT:TR:00500	Altitude Sensor Update Rate	The altitude sensor shall update the altitude every 10 s.	This conforms with UCD:TR:00200 and supports experimental triggering within 500 metres of the desired altitude mark.

3.3.2.5 Communications

Table 12: Mission Specific Communications Requirements

Requirement ID	Title	Description	Justification
SQT:TR:00600	Downlink Capacity	The communications architecture shall be capable of transmitting all required data packets in a timely manner	This is to ensure the acquisition of secondary and tertiary mission validation figures.

3.3.2.6 Triggering Algorithms

Table 13: Triggering Algorithms Requirements

Requirement ID	Title	Description	Justification
SQT:TR:00700	Pressure Trigger	The pressure (Good) trigger will activate for ambient pressures of less than 35 mbar	This requirement ensures that the exposure mechanism operates in suitable conditions before balloon failure.
SQT:TR:00710	Altitude Trigger	The altitude (Bad) trigger shall activate at a 23 km altitude provided that the pressure data is inconsistent with predictions	This requirement ensures that the exposure mechanism operates before the TupperSat reaches balloon bursting altitude.
SQT:TR:00720	Falling Trigger Operation	The falling (Ugly) trigger shall activate if telemetry data shows that the TupperSat has been falling for a significant amount of time (10 s).	This requirement forces exposure mechanism operation if the balloon has burst. Acts as failsafe to retrieve tertiary mission validation data (servo/trigger flags).
SQT:TR:00730	Falling Trigger Minimum Altitude	The falling trigger shall not activate below an altitude of 2 km.	This is to ensure that trigger isn't activated during TupperSat setup on launch day.

4 Function Tree

The function tree in Figure 1 details the processes that will need to be carried out in order for the mission objectives to be met. This function tree will directly influence the product tree, and the interface control approach.

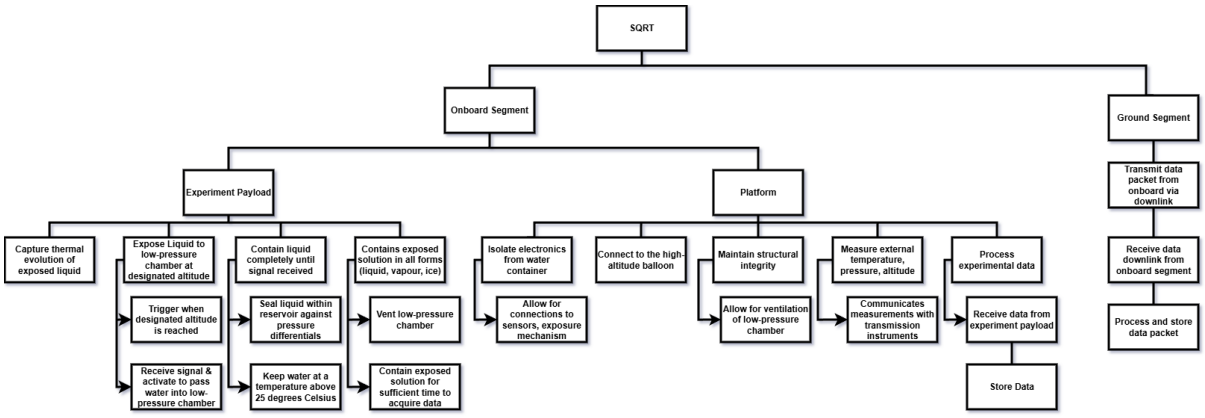


Figure 1: SQRT Function Tree

5 Product Tree

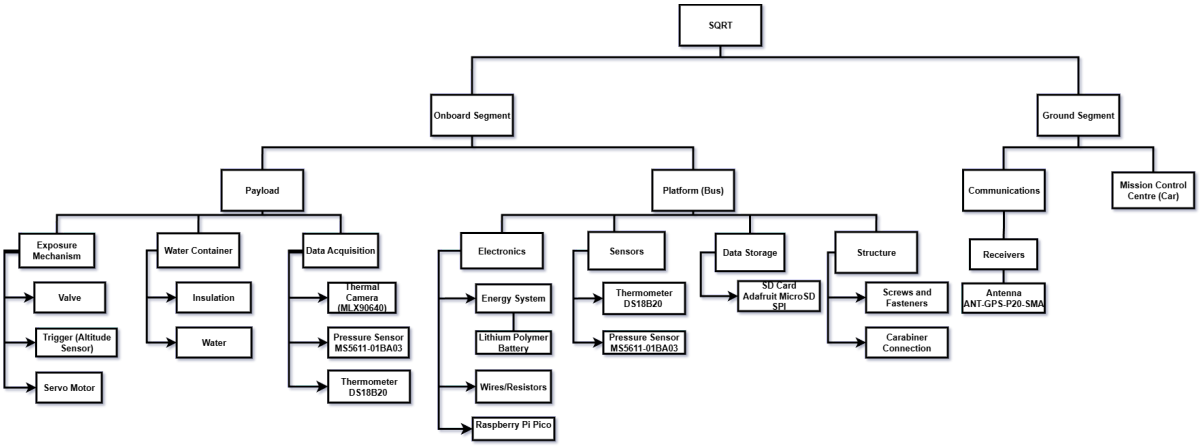


Figure 2: SQRT Product Tree

The product tree shown in Figure 2 has been updated to include the thermal sensor (MLX90640). The Raspberry Pi Pico, a notable omission from the PDR document, has also been added. It outlines the resources and consumables that are needed to perform the processes in the function tree. It also provides a useful visual aid for the ongoing procurement process, ensuring that all prototyping and subsequent testing can be performed in a timely manner.

6 Interface Control

The interface control plan is concerned with the smooth cooperation of subsystems within the TupperSat during the flight campaign. This includes the shared resource pool that the subsystems will be pulling from, as well as the outputs that different subsystems will be producing. The interfaces will be separated into internal and external sets. The internal interfaces will describe connections within the payload e.g. sensor => Pico, Pico => SD card. Interfacing internal structures will include the connection of different subsystems to the power source via the Pico and the delegation of commands to different sensors via the Raspberry Pico. External interfaces are related to how the TupperSat architecture will interact with the external environment (ground segment, ambient conditions, High-altitude balloon). Controlling these interfaces will involve minimising the loss of downlinked data packets and ensuring that the TupperSat maintains structural integrity throughout the flight campaign.

6.1 Internal Interfaces

Raspberry Pico
The Raspberry Pico will be the primary instrument used to implement interface control for internal subsystems. It will be responsible for the distribution of power and the subsequent storage of experimental/telemetry data. Subsystems interfacing with the Raspberry Pico can be seen in the table below.

Table 14: Raspberry Pico Interfaces

Subsystem	Interface Type	Inputs	Outputs
Power Supply (Lithium Polymer Power Bank)	Hardware	Insulated, waterproof Environment	Power
Pressure Sensor	Software/Hardware	Power, I^2C connection	Timestamped Pressure & Temperature values
Thermal Sensor	Software/Hardware	Power, Device Address & Port, I^2C connection	768 raw values & calibration constants for processing.
Servo Motor	Hardware	Activation signal, Power	Actuation of Valve to release tube and pass water sample to low-pressure chamber
SD Card Module	Software/Hardware	SPI connection, formatted data, data handling pipeline	Formatted datafiles
Temperature Sensor	Software/Hardware	1-wire connection	Internal temperature values in valve proximity.
Comms (T^3)	Software/Hardware	UART connection, formatted data packets for downlinking, commands for scheduled transmissions	Ground-bound transmissions for communication with ground segment
Triggering Algorithm	Software	Files containing Time, Pressure and Altitude Values	Tiered flagging based on input values (Good: Pressure threshold, Bad: Altitude threshold, Ugly: Falling (consistent loss of altitude))

Power Supply: The power supply will be required to supply power to all subsystems via the Raspberry Pico. Power will be delivered to the Pico through a micro-USB connection. It is important that all interfacing subsystems are verified to be compatible with the 3.3 V voltage supplied by the Raspberry Pico.

Total mission power consumption must also be considered over the expected 6 hour flight duration. In particular, the thermal sensor will be continuously operating at a refresh rate of 32 Hz, contributing to a sustained power draw. As well as this, peak current demands, which will be associated with radio transmissions and servo actuation, will have to be accounted for to prevent voltage drops. Power budgets have been drafted, confirming that the battery capacity is sufficient to support nominal and peak loads for the mission duration.

Pressure Sensor: The pressure sensor will interface with the Raspberry Pico via the I^2C bus and will require 4 pins (2 GPIO). Pull-up resistors may also be required for the data and clock lines, depending on the sensor configuration. Pressure sensor values are prioritised within the onboard data storage hierarchy, as the values will be required for determining the trigger flag status at a given instant. The pressure and temperature measurements produced by the pressure sensor will be stored in suitably named data files. Values will be read in from the sensor on a sampling rate of approx. 1 Hz, providing sufficient temporal resolution for trigger condition monitoring and satisfying requirements, while minimising unnecessary power consumption.

Thermal Sensor: The thermal sensor will be continuously taking measurements for the duration of the mission (6 hours). The sensor will be configured via its inbuilt control registers to operate at 32 Hz refresh rate. The sensor will interface with the Raspberry Pico via the I^2C bus and will require 4 pins (2 GPIO).

During each loop, 32 frames (32x24 arrays) will be read in from the sensor's internal RAM. The raw values read in will be calibrated by the Raspberry Pico before being stored in appropriately named and formatted data files on the SD card. The outputted thermal will be prioritised as the most valuable acquisition of the mission, giving it preference in any writing conflicts.

Servo-Motor: The servo motor is a critical component of the exposure mechanism subsystem and is responsible for actuating the pinch valve. This physical interaction enforces that the valve must be mechanically compatible with the servo motor, avoiding any structural vulnerabilities. In order to conform to these interfacing constraints, a number of valve options were considered and evaluated in SQT-CDR-HDF-0. The valve chosen (Pinch valve) provides good sealing and high tolerances to temperature and pressure extremities.

Electrically, the servo motor interfaces with the Raspberry Pico via the trigger status signal, which determines when the motor is activated. As this is a single-use operation, control software must prevent repeated activation and deactivation of the motor (opening and closing of the valve). The mechanical design of the subsystem will have to ensure that the valve is active-open, minimising continuous power draw during operation.

SD Card Module: The SD card's sole interface is with the Raspberry Pico. The data storage policy is enacted via a script on the Raspberry Pico which will be continuously called in each loop of the mission software pipeline to store the relevant telemetry and science data. Careful consideration will have to be given to the wiring process to ensure that MOSI and MISO are not misaligned. Further details on the SD card interfacing and data management strategy can be found in SQT-CDR-DMP-0.

Comms (T^3): The communications suite onboard the TupperSat will be required to interface both internally with the Raspberry Pico and externally with the ground segment receiver antenna. Controlling the internal interface between the Pico and the downlink architecture will be carried out by the main mission pipeline. Timers will be used to schedule transmission of telemetry/science data packets to ground. As detailed in the software development plan, the science data timers will be checked first to ensure that packets are being sent consistently, by the minute. To populate the data packets, the Pico will access the data files in the SD card, read in the most recent measurements, and format the data packet. The refined data packet will then be transmitted to ground utilising the TupperSat Telemetry Thingmabob (T^3). This instrument will physically interface with the Raspberry Pico's TTL serial port. The serial pinout used will consist of a ground, power source, transmission and receiving.

6.2 External Interfaces

The external interfaces will determine how the TupperSat will interact with its surroundings and with mission control. Controlling these interfaces will involve considered hardware design and the design of low-cost data packets that can be tracked back to the SQRT mission at a given timeframe. The external interfaces are summarised in the table below.

Table 15: External Interfaces

Interface	Summary	Solution
Ground Segment	Interfacing with mission control is required to fulfill customer requirements, transmit experimental data, and facilitate recovery operations.	Ensure proper telemetry link, antenna orientation, and RF compatibility.
Ambient Environment (Temperature)	Operating temperatures for electronics (sensors, Pico, GPS) must be maintained under external temperatures as low as $-50°C$. Water container must exceed 25°C for proper flash evaporation.	Use insulation and passive heating from electronics. Design sealed dividers between electronics bus and low-pressure chamber.
Ambient Environment (Pressure)	Atmospheric pressure drops to ≈34 mbar at 23 km altitude; low pressure affects valve performance. The water container will need to be kept at near 1 atm in order to maintain pressure differential required for flash evaporation.	Verify valve actuation and sensor functionality at low pressure. Ensure that valve holds water in the container against pressure differentials up to 950 mbar.
Ambient Environment (Moisture / Humidity)	Moisture may condense on electronics at high altitude.	Design will ensure that holes for venting/structure will be isolated from critical components, sensors and electronics.
High-Altitude Balloon Connector	TupperSat exterior infrastructure needs to support stable, structurally sound connection to the High-Altitude Balloon.	Design will ensure that eye nuts are positioned appropriately to prevent catastrophic structural failure.

T^3 to Ground Segment: This interface represents the primary external communication link between the TupperSat and mission control. Controlling this interface will facilitate the retrieval of secondary and tertiary mission validation criteria. Provided that internal interfacing is controlled adequately, transmissions will be sent out at a rate of 2-3 packets per minute. The positioning of the transmitting dipole antenna on the underside of the TupperSat will have to meet loss requirements and avoid structural interference from the TupperSat's external mechanical architecture. Antenna performance and overall communication range will be validated during the long-range flight test to ensure compliance with technical communication requirements.

7 System Development Plan

Upon completion and acceptance of the Critical Design Review, and the proposed structure therein, work will commence on the development of the mission architecture (hardware, software). Prototyping of key subsystems will continue, using the resources that are currently available in-house. The software suites required to interact with these subsystems and reliably store their outputted data will be developed.

Prototyping/Procurement: As per SQT-CDR-PMP-0, phase 3 will consist of the finalising of material procurement and the assembly of prototyping for tests of function. This testing process will outline adaptations that need to be made before final assembly and subsystem integration. Stress tests will highlight any potential structural weaknesses that will need to be addressed before the prototypes can be integrated into the holistic flight model for validation testing.

Software: Software development will run alongside the physical assembly. The software will attribute a class object to each sensor, describing their designed dependencies and outputs. Software work packages will be assigned to different team members, who will produce pipelines for retrieving and calibrating raw

data. Once the subsystem software scripts have been produced and their outputs formatted, a combined pipeline will be constructed to facilitate the smooth interfacing of all onboard systems. The data storage policy outlined in the DMP will also materialise in the form of a data storage sequencing script. This will prevent the occurrence of file corruption or loss of data. The flight-ready pipeline will be tested alongside the sensors to ensure that it can handle the outputs and store them safely, avoiding all conflicts.

Flight Tests: The latter stages of phase 3 will revolve around the flight tests detailed in SQT-CDR-FTSTP-0. These tests will verify flight readiness and highlight any potential failure modes that will need to be promptly remedied before the launch window opens. The long range test will appraise the strength of the signals transmitted by the model, and the wholeness of the data packets. The conclusion of phase 3 will coincide with the Flight Acceptance Review (FAR) which will appraise the TupperSat's flight readiness and, if successful, clear the flight model for launch.